

Intro to Biology

Mr. Queenan – Biology 9

Unit 1.2

For more information, see text Ch 1.1 (p 1-10) & Ch.1.3 (p 22-30)

What is:

- Science?
 - the state of knowing : knowledge as distinguished from ignorance or misunderstanding
 - Merriam Webster - <https://www.merriam-webster.com/dictionary/science>
- Biology?
 - The scientific study of life..... OR
 - The state of knowing or understanding LIFE
- All forms of life share common properties

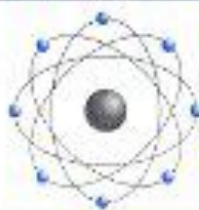
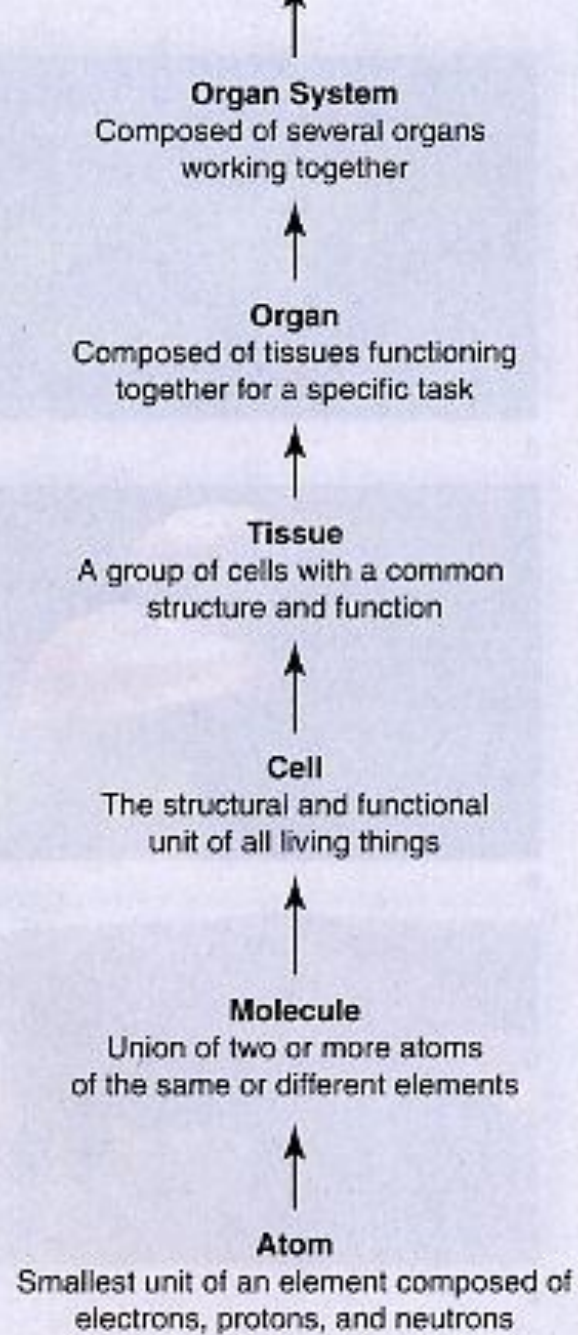
Characteristics of Life

- Life is extremely diverse
- How do we begin to categorize things as living or non-living?
- Basic Characteristics (7) shared by all living things

Characteristics of Life

1. All living things are organized

- Start with basic units of organization, and move up in complexity
- Can start from small to large or large to small but, there is an important reason to think from small to large
- As you move up the hierarchy in complexity, new ***EMERGENT PROPERTIES*** develop



**Biosphere**

Regions of the Earth's crust, waters, and atmosphere inhabited by living things

**Ecosystem**

A community plus the physical environment

**Community**

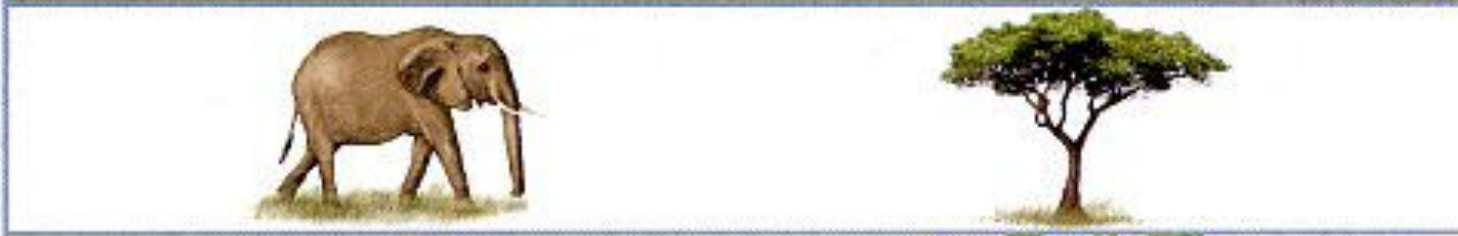
Interacting populations in a particular area

**Population**

Organisms of the same species in a particular area

**Organism**

An individual; complex individuals contain organ systems

**Organ System**

Characteristics of Life

2. Living things acquire materials and energy

- Living things require outside sources of energy and/or nutrients to carry on life
 - What is the main source of energy for most life on Earth?
 - **The Sun**
- Organisms are able to perform chemical reactions to metabolize energy
- Ex include Photosynthesis or cellular respiration
 - Will discuss in more detail in future units

Characteristics of Life

3. Living things maintain homeostasis

- Living organisms must maintain a state of biological balance to survive
- There is a tolerance range for survival
- These biological adjustments do not require conscious activity – controlled mostly by nervous systems
- Examples:
 - Body temperature and sweat
 - Blood sugar level regulation
 - pH

Characteristics of Life

4. Living things respond

- Interact with environment
- Interact with other living things
- Examples
 - Locomotion towards or away from chemical in unicellular organisms
 - Migration of birds based on season
 - Plants leaves moving towards the sun
 - <https://youtu.be/zol3174S8Xg>

Characteristics of Life

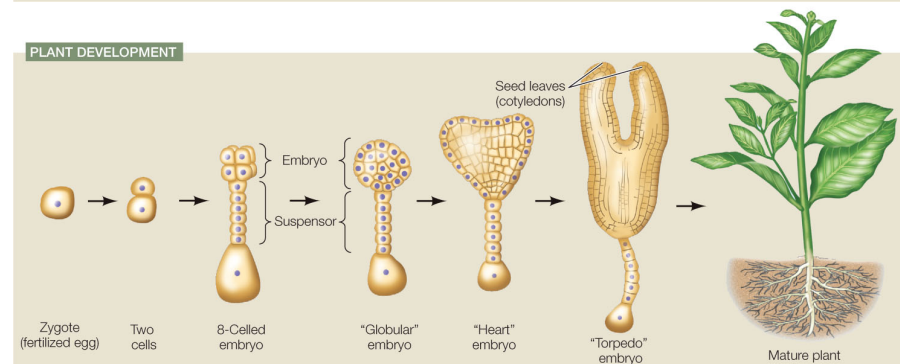
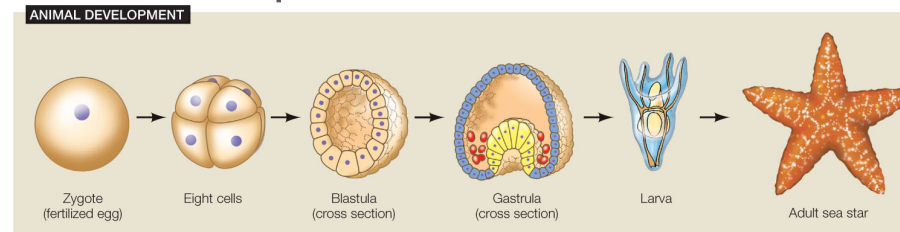
5. Living things can reproduce

- Life *only* develops from life
- Therefore, living things must be able to create new organisms, and pass on genetic material, or **genes**
 - Unicellular splitting or division
 - Asexual
 - Identical to parent
 - Union of sperm and egg
 - Sexual
 - Combination of parents

Characteristics of Life

6. Living things can develop

- After reproducing, an organism must go through a set of stages to develop into a mature organism
 - Can increase in size
 - Can become more complex
 - Can change over time
- The organisms genes include not only the material involved in reproduction, but also the blueprint for an organisms development!



Characteristics of Life

7. Living things have adaptations

- Modifications that make organisms able to better function in a particular environment
- These adaptations have led to the diversity of life on Earth
- These adaptations to environments over time also have another name
- **Evolution**
- All living things have the capacity to evolve
- Allows a way for organisms to persist, despite a changing environment

Characteristics of Life

Living things:

1. are organized
2. acquire material and energy
3. maintain homeostasis
4. respond
5. can reproduce
6. can develop
7. have adaptations

These 7 characteristics drive every unit of study we will explore this year!

Evolution, genetics, ecology, cytology, etc!

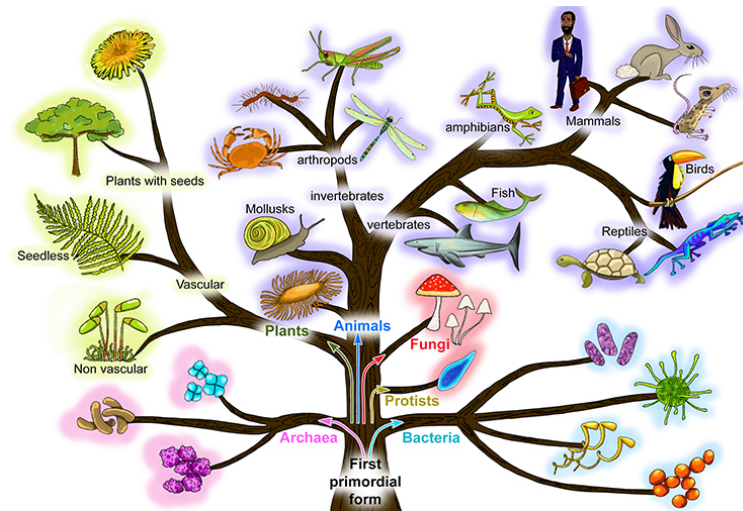
Activity

- What factors could you use to justify the following objects are alive?
- What factors would you use to show they are not?

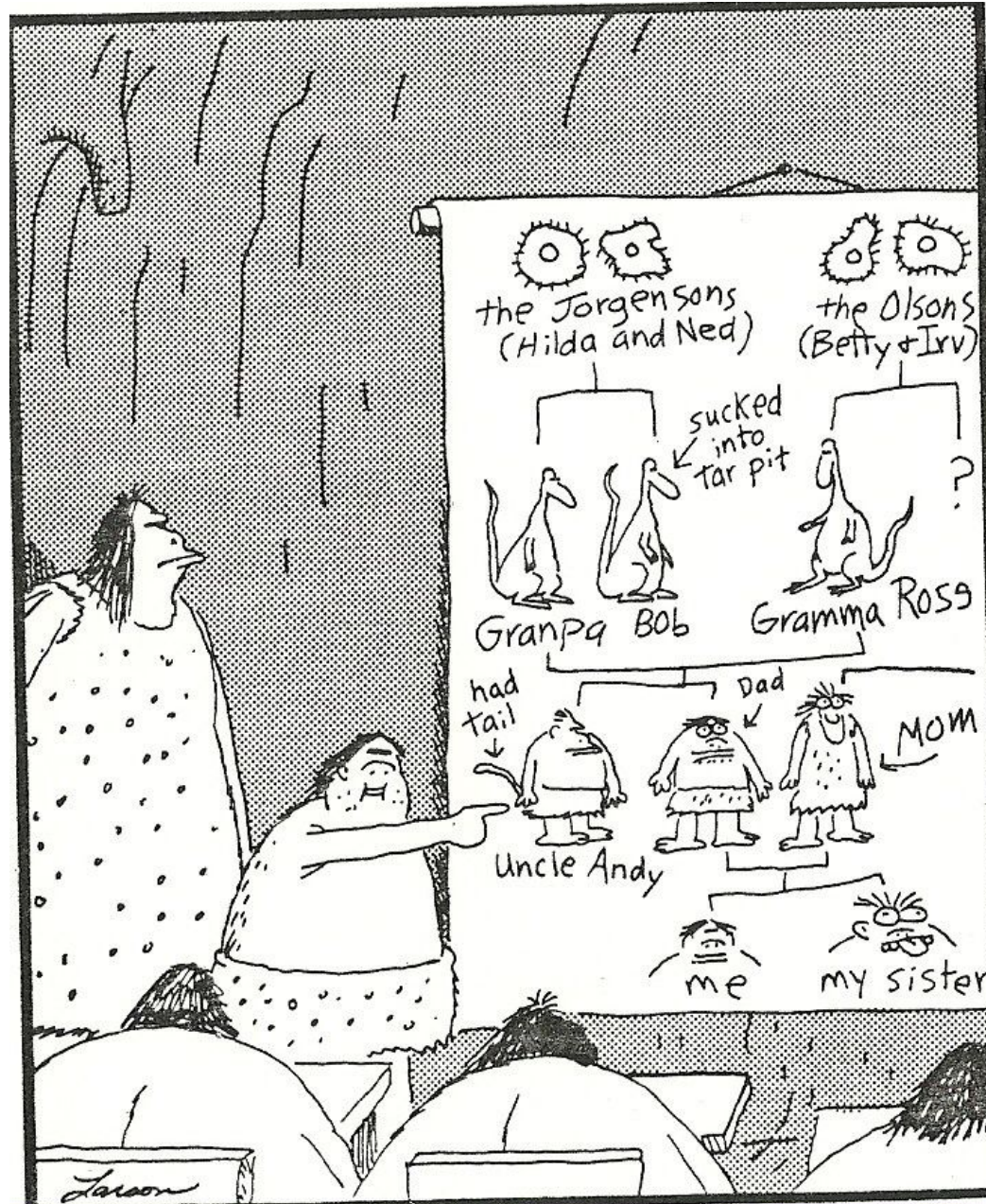


Organizing Diversity

- Categories are needed to make sense of all of Earth's diversity
 - Scientists have identified close to 2 million different species
 - Actual estimates of Earth's diversity are 10-100 million species
- **Taxonomy** is the discipline of identifying and grouping organisms
 - Names species
 - Classifies species into a hierarchy of broader groups
- **Systematics** is the study of evolutionary relationships between organisms

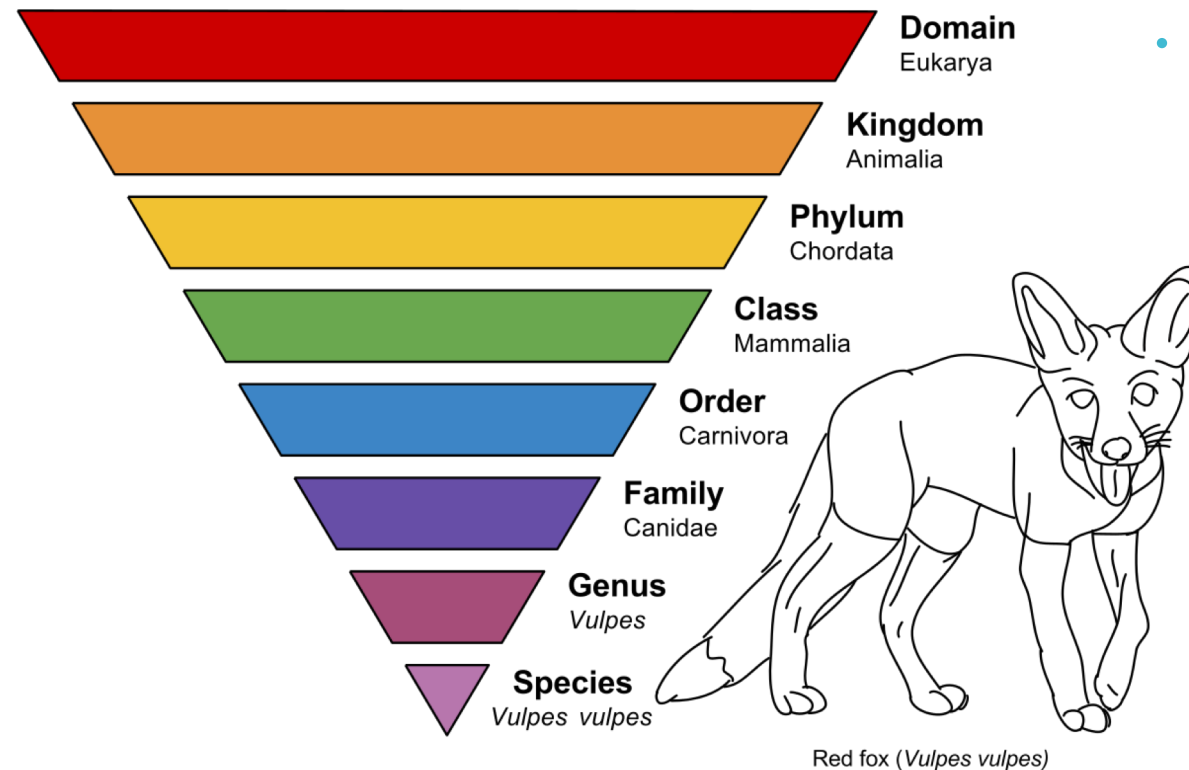


Organizing Diversity



Organizing Diversity

- Basic classification categories go from least inclusive to most inclusive (just like levels of organization discussed earlier)
- Least inclusive is **species**, and is a group of interbreeding individuals
- As you move up a level in hierarchy, the level contains more individuals, but the individuals share less characteristics



- 3 Domains
 - Bacteria
 - Archaea
 - Both Prokaryotes
- Eukarya
 - 4 Kingdoms

Organizing Diversity

- Domain Bacteria
 - are the most diverse and widespread **prokaryotes**.
 - Prokaryotes are organisms lacking membrane bound nucleus
- Domain Archaea
 - are prokaryotes that often live in Earth's extreme environments.
 - Deep sea vents, volcanic chambers, etc
- Domain Eukarya
 - All organisms are **eukaryotes**, or contain a membrane bound nucleus
 - Range from single-celled protists to multicellular organisms

Organizing Diversity

Table 2: A Six-Kingdom System of Classification

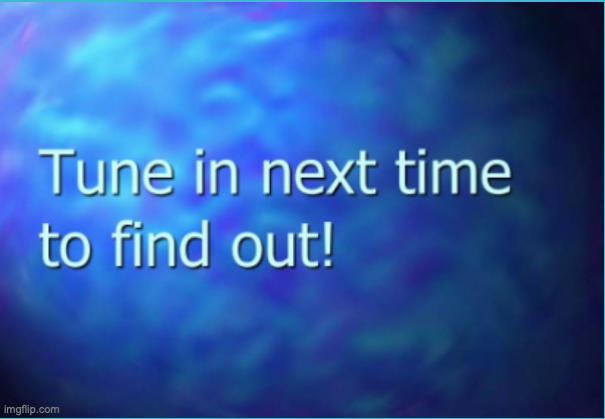
	Kingdom	General characteristics	Cell wall	Representative organisms
	1. Eubacteria	• simple organisms lacking nuclei (prokaryotic) • either heterotrophs or autotrophs • all can reproduce asexually • live nearly everywhere	often present (contains peptidoglycan)	bacteria, cyanobacteria
	2. Archaeobacteria	• prokaryotic • heterotrophs • live in salt lakes, hot springs, animal guts	present (does not contain peptidoglycan)	methanogens, extreme thermophiles, extreme halophiles
	3. Protista	• most are single celled; some are multicellular organisms; some are eukaryotic • some are autotrophs, some heterotrophs, some both • reproduce sexually and asexually • live in aquatic or moist habitats	absent	algae, protozoa
	4. Fungi	• most are multicellular • all are heterotrophs • reproduce sexually and asexually • most are terrestrial	present	mushrooms, yeasts, bread moulds
	5. Plantae	• all are multicellular • all are autotrophs • reproduce sexually and asexually • most are terrestrial	present	mosses, ferns, conifers, flowering plants
	6. Animalia	• all are multicellular • all are heterotrophs • most reproduce sexually • live in terrestrial and aquatic habitats	absent	sponges, worms, lobsters, starfish, humans

<http://3ubio.weebly.com/taxonomy.html>

Organizing Diversity

- Diversity is observed across all levels of biological organization
 - From cell structure, to organisms, to populations, to ecosystems and biodiversity.
 - We will study the interplay of each of these over the course of the next few months, moving from the smallest (Biochemistry and cytology) to the largest (biomes and ecology)

Coming Soon in Unit 1



Tune in next time
to find out!

- The Process of Science
 - The Scientific Method
 - Graded activity
- Science Technology
 - The Microscope
 - Graded activity
- Biotechnology, Bioengineering, & Bioethics
- Journal Club
- Objectives Submission & Review
 - Unit Assessment